

# Bayesian Networks: A Decision Tool to Improve Portfolio Risk Analysis

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*This paper demonstrates how Bayesian networks, a graphical modeling tool, can improve analysts' forecasts, portfolio decision-making, and risk analysis. Bayesian networks combine historical quantitative information with qualitative information in a systematic way. Findings in behavioral finance show buying and selling investment behavior that indicates biased decision-making, but most behavioral finance literature is descriptive, not normative. Our goal is to improve rational financial decision-making by helping analysts eliminate bias from their probability assessments and systematically improve value-at-risk forecasts. [G11, G19, C4]*

*Portfolio management is a very special problem in engineering, of determining the most reliable and efficient way of reaching a specified goal, given a set of policy constraints, and working within a remarkably uncertain, probabilistic, always changing world of partial information and misinformation, all filtered through the inexact prism of human interpretation (Ellis 1985, p. 53).*

■ Security analysts evaluate a variety of information to decide whether to buy, sell, or hold a security. Research on security analysis concentrates primarily on two different areas. The first is pricing and valuation models. The second is the relation between firm or economic variables and earnings forecasts. We use a Bayesian network to model economic relations to produce earnings forecast for each stock in a portfolio and a return distribution for the portfolio. The output of the model is a probability distribution that combines historical information with current news.

Traditional pricing models such as the capital asset pricing model (CAPM) or arbitrage pricing theory (APT) describe how economic variables and firm characteristics are related to stock returns. The models are based on historical and quantitative data, and the results are averages for a typical firm. Most analysts use this historical quantitative analysis as part of an overall approach that also includes a wider variety of information. Analysts typically concentrate on special situations and individual cases, not on the average. Their information includes historical data and qualitative, imprecise evidence that may affect a firm.

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An analyst may consider, for example, how effective or trustworthy a firm's management is. Or, what is the effect of China's entry into the World Trade Organization on a particular line of business? How reliable are a firm's financial statements? We show how to integrate this type of information with historical quantitative data using a graphical decision-modeling tool, Bayesian networks.

In portfolio management, analysts must assess a large amount of sometimes conflicting data to make a decision based on uncertain information. We suggest that Bayesian networks are especially well suited for this task. They help experts represent uncertain, ambiguous, or incomplete knowledge that portfolio managers and analysts often deal with.

The output of the model is a probability distribution of portfolio value. As the entire distribution is modeled, measures of risk and value-at-risk (VAR) are modeled naturally. When new evidence is added its effects on other variables in the model and on risk are also computed.

There are two types of inputs to the model – the graphical relation, and a set of equations and conditional probability distributions described by the graph. The conditional probability relations can be estimated from historical data or from expert judgment. Shenoy and Shenoy (2000) show how traditional expected return models such as the CAPM and APT can be used to model relations in a Bayesian network.

Here we show how to combine quantitative data with qualitative or soft information in a systematic way. Our example demonstrates how to combine macroeconomic factors and firm-specific factors, but the methodology is flexible enough to reflect an individual analyst's decision-making process.

This work makes several contributions. First, Bayesian networks have been used in different decision support system contexts that combine qualitative and quantitative information.<sup>1</sup> We suggest they be used as a compact and intuitive decision-modeling tool to represent different aspects of uncertainty inherent in security analysis. Second, we provide a methodology to combine economic and firm-specific variables. The example includes a range of factors that determine stock returns at the macroeconomic, industry, and firm levels. Third and most important, we illustrate a systematic way to add new information (or evidence) to current security analysis techniques and to see its effect on valuation as well as portfolio risk. This feature of Bayesian networks helps analysts explicitly model new information. Finally, the framework we present can be used to create an expert system where learning takes place as part of a neural network.

## I. Security Analysis and Decision Making Under Uncertainty

Behavioral finance research has shown that some aggregate market-pricing patterns are consistent with human cognitive patterns that are considered "irrational" or non-value-maximizing. Hirshleifer (2001) summarizes a range of behavioral finance articles and finds that "expected returns are determined by both risk and *misvaluation*." He attributes this misvaluation to various cognitive and information processing biases. Similarly, Alpert and Raiffa (1982) demonstrate most people consistently underestimate standard deviations in probability distributions.

Traditional expected return models fail to capture the subjective and inexact nature of probability distributions associated with the economic variables in a financial model. We know that financial analysts operate in a world described by approximate and subjective probabilities that describe the relations among variables in the model. An automated Bayesian network decision support system makes it possible to capture the probability judgments analysts make over time, compare them to actual data when they become available, and then provide feedback on a timely basis. Hagstrom (1999) observes:

*Whether or not they recognize it, virtually all decisions that investors make are exercises in probability. For them to succeed, it is critical that their probability statement combines the historical record with the most recent data available. And that is Bayesian analysis in action (p. 119).*

A Bayesian network is a network of relations. It reflects the states of some part of a world that is being

modeled, and describes how those states are related by probabilities. Studies have shown that approximate probabilities, even subjective ones that are guessed at, give very good results, implying that Bayesian networks are robust to imperfect knowledge.<sup>2</sup> Since a Bayesian network model reflects the analyst's understanding of a particular set of relations, it is an ideal tool to help experts represent uncertain, ambiguous, or incomplete knowledge associated with security analysis.

Applications of Bayesian networks in different contexts including credit risk modeling and auditing indicate that the combination of several pieces of imperfect knowledge can yield strong conclusions. The use of Bayes' rule also allows the analyst to efficiently update all probabilities when any one piece of information changes.

A Bayesian network consists of two parts — a qualitative, graphical structure of the relations in a model, and a quantitative structure represented by the probability distributions indicated by the graph. In general, historical information can provide a framework or baseline model to develop prior distributions. New quantitative information, qualitative information, or evidence can be added to the network as appropriate to develop posterior probabilities. We propose a flexible decision support system that provides a tool to improve individual decision making, to facilitate learning, and to enhance performance. The network provides a simple way to incorporate probabilities into the decision maker's analysis.

Decision makers in many different contexts combine quantitative data and qualitative information, and Bayesian networks have been applied in a wide variety of decision-making uses. Some examples are venture capital financing (Kemmerer, Mishra, and Shenoy, 2001), auditing (Gillett and Srivastava, 2000), medical diagnosis (Bielza, Ríos-Insua, and Gómez, 1999), and software design (Horvitz, Breese, Heckerman, Hovel, and Rommelse, 1998).

There are three important steps in building a Bayesian network — the graphical representation of conditional probability distributions, specification of the numerical relations among the variables, and making inferences or decisions based on new evidence. Several researchers have shown that the graphical representation may be the most critical. Henrion, Provan, Favero, and Sanders, (1994) and Darwiche and Goldszmidt (1994) show that results are more sensitive to the qualitative structure of the model and robust with respect to the numerical specification. Generally, it is better to have a very good graphical representation of the decision problem with approximately correct probability distributions than to have a poor graphical model with very precise probability distributions.

<sup>1</sup>For several examples, see Jensen (1996).

<sup>2</sup>See Jensen (1996) for a review.

Several commercial software tools are used to estimate Bayesian networks. They allow the user to draw the graph and specify the numerical relations among the variables, and the software calculates the inferences based on these inputs. The inference results are shown graphically as probability distributions for the network. Hugin ([www.hugin.com](http://www.hugin.com)) and Netica ([www.norsys.com](http://www.norsys.com)) can both be used to automate the process. Netica is used for the present example.

## II. Developing a Bayesian Network for Security Analysis

We assume the analyst is using a top-down approach, starting with economic and industry analysis before proceeding to a firm analysis.

We describe the three basic steps in the Bayesian network construction. The first step is developing the graphical model, which includes identifying the relevant variables and specifying whether they are independent or not. The second step is specification of the numerical relations among the variables that are not independent. The third involves updating the model with new evidence.

### A. Graphical Representation

The graphical model is a directed, acyclic graph where nodes represent variables, and directed arcs (arrows) represent the conditional probability relations assumed in the model. The variables and the arcs between the variables are the inputs to the graph. The diagram reflects an analyst's understanding of the relations.

Exhibit 1 shows a general graphical structure of an  $n$ -stock portfolio that models each stock in the portfolio separately with respect to the economic factors that may affect it. This model is a standard representation of the top-down approach, where economic factors affect industry factors, which in turn affect a particular stock's return.

The first step in a general top-down security analysis is to identify the specific economic, industry, and firm-specific variables that make up the Bayesian network. An analyst usually starts with a number of specific variables already identified. Academic research can also provide some examples. The CAPM is a very simple example of an economic and a firm variable – the market return and a security return.

The shaded nodes in Exhibit 1 represent groups of variables. In a complicated model, it is useful to group variables and relations for a better overview of the structure. In this model, multiple economic factors affect multiple industry factors. For example, the arc from *Economic Factors* to *Industry 1 Factors* indicates that the probability distribution of the industry factors is conditioned on the economic factors.

The arcs leading to the stocks indicate that the stock return distributions are conditioned on industry factors and stock-specific factors. The arcs also indicate which type of probability distribution must be estimated. A node with arcs leading out only indicates a marginal probability distribution. A node with arcs leading into it indicates a conditional relation.

The absence of a directed arc from a node is also meaningful, because the absence indicates conditional independence. For example, there are directed arcs leading from each of the stocks to the portfolio nodes, but not from any economic, industry, or firm-specific factor, because this particular network assumes that the portfolio distribution is independent of all other factors, given the distributions of individual stock returns.

One major advantage of a Bayesian network is that it can effectively represent conditional probabilities. In Exhibit 1, for example, the joint probability distribution for the variables *Economic Factors* ( $EF$ ), *Industry 1 Factors* ( $IF1$ ), *Stock 1 Factors* ( $SF1$ ), and *Stock 1* return ( $SI$ ) is represented as

$$P(EF) \times P(IF1|EF) \times P(SF1|IF1) \times P(SI|SF1) \quad (1)$$

where  $P(\cdot|\cdot)$  denotes the conditional probability distribution. This representation indicates that the return on Stock 1 is independent of the industry factors as well as the economic factors, given information on the state of Stock 1 factors.

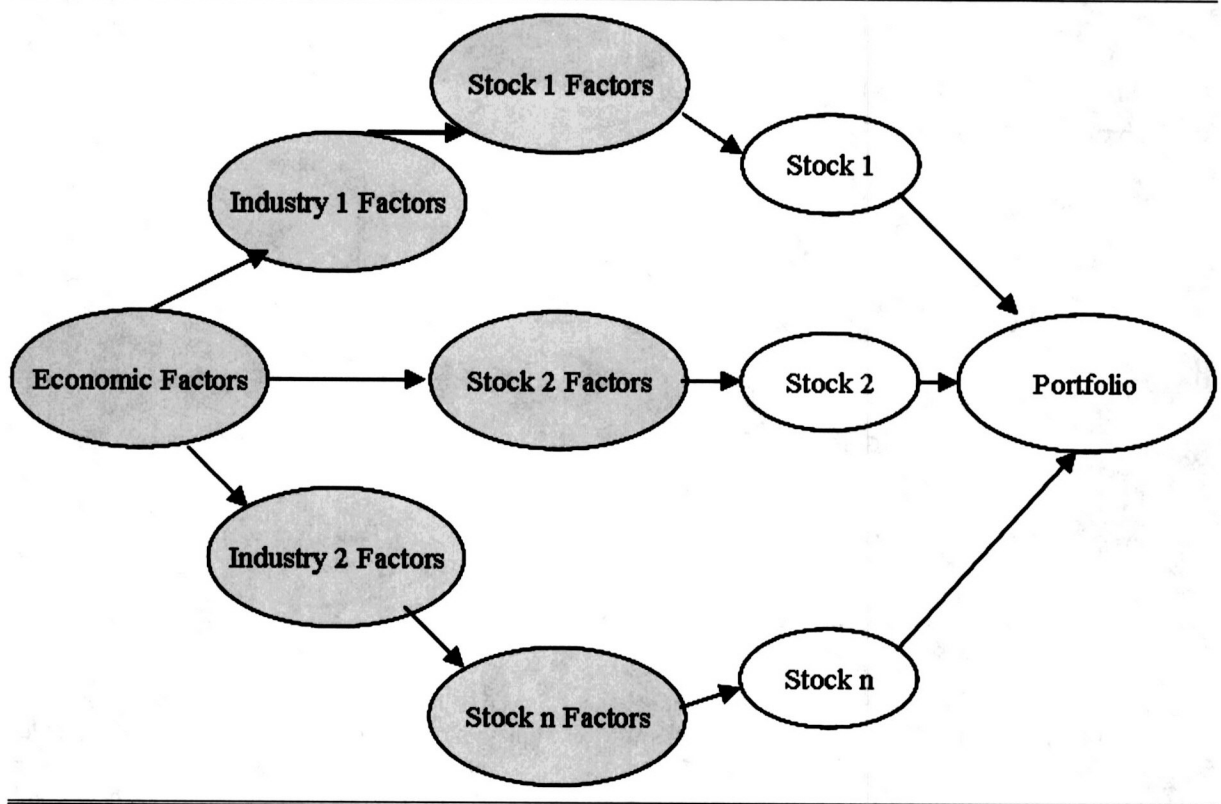
The use of conditional independence arguments in the modeling phase not only helps the analyst to better understand the relations among variables, but also makes it easy to update all probability distributions and associated risk measures, given evidence on certain variables in the model, an issue especially important if the model is updated automatically.

Suppose an analyst wants to forecast pretax income using a pro forma income statement. The relations among the income statement variables may be graphically represented in Exhibit 2.

The diagram in Exhibit 2 suggests that the decision maker assumes that revenue and cost of goods sold (COGS) for each segment are independent, but that the revenue and cost of goods sold for each segment are related to the revenue for that segment. This representation also assumes that operating expenses are independent of each segment and other income and expenses.

LSB Industries is a good example of the situation shown in Exhibit 2. LSB manufactures chemicals and climate control products, and provides industrial engineering services. Each of these segments of its business operates in a different market.

Exhibit 2 is a less appropriate model for Limited Brands. Limited Brands classifies its two main segments of business as apparel and intimate brands. It attempts to locate stores in the two lines of business in the same

**Exhibit 1. General Graphical Portfolio Model**

mall, and the target market overlaps in many cases. Therefore, in the Limited Brands case, the revenue from each segment is not independent. Exhibit 3 shows a graphical representation of the case of Limited Brands.

The analyst benefits in several ways from the graphical construction of a Bayesian network. First, construction of the diagrams helps the analyst clarify and the view of the relations among the variables. Second, the analyst may better understand how a decision is reached. He or she may fully understand which variables are used, but not necessarily how the variables are combined and weighted to come up with a decision.

Shenoy (2005) finds that while almost all analysts agree that qualitative information is important, when they are questioned about how it was incorporated in a decision, most analysts could not be specific.

## **B. Determining Numerical Links Between Variables**

Each node in the Bayesian network is a variable that is described either as a constant value, a probability distribution, or as a function of other variables. In a Bayesian network, the primary focus is on determining the probability distribution of the relevant nodes. In spreadsheet

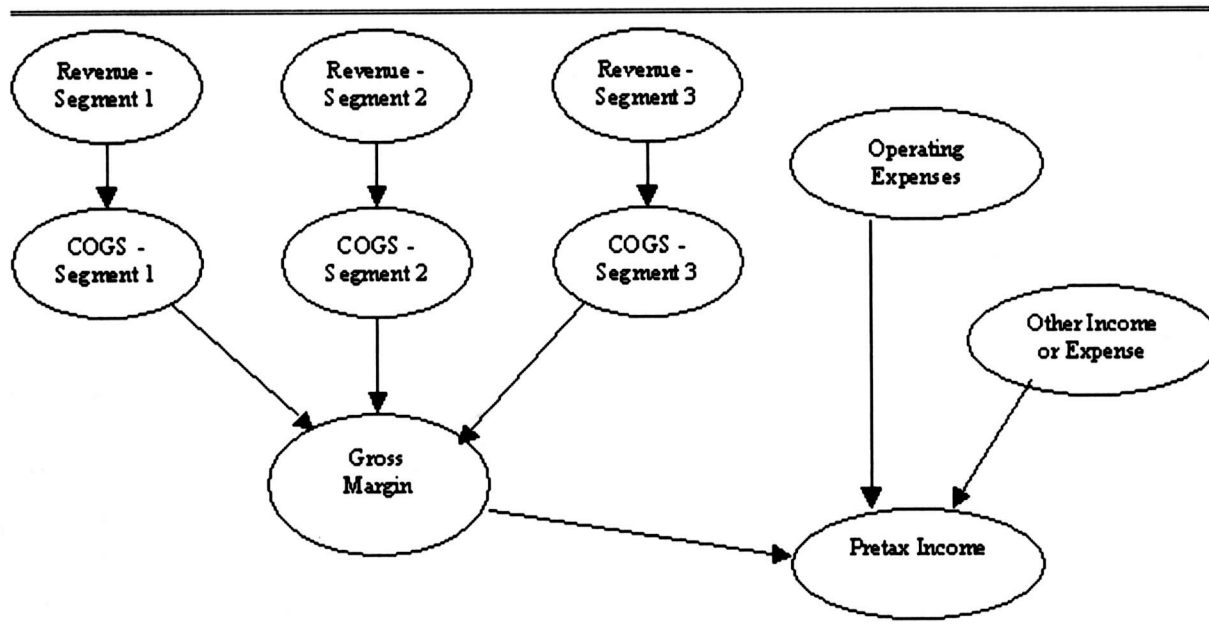
analysis, the primary focus is on determining variable values and the functional relations among the variables. Specialized software tools for spreadsheets, such as @Risk and Crystal Ball, also combine constant functional relations and probability distributions using simulation.

Bayesian networks use an efficient algorithm called local computation to find the probability distribution of interest. Local computation leads to fast updates of probabilities and is especially important when a large number of macro- and microeconomic variables are involved. New information can easily be incorporated to update the entire network.

There are two primary ways to find probability distributions for the nodes in the network. One is to use historical data, and the other is to use subjective probability judgments. The two methods can also be combined.

There are also two types of probability distributions in a Bayesian network – marginal distributions, and conditional distributions. A marginal distribution for a node is one that does not have an arrow leading into it. For example, in Exhibit 3, *Revenue-Segment 1*, *Operating Expenses*, and *Other Income or Expense* variables are all marginal probability distributions. In Exhibit 2 there are more marginal distributions – one for each segment's revenue and cost of goods sold, and also for *Operating*

## Exhibit 2. Example Model for Pretax Income



*Expenses*, and *Other Income or Expense*. A conditional probability distribution is indicated by a node that has at least one arrow leading into it. In Exhibit 3, COGS for both segments are conditional distributions, as are *Gross Margin* and *Pretax Income*.

Regression analysis is a familiar way to use historical data to find a conditional probability distribution. The dependent variable in a normal form regression is normally distributed with the mean estimated by the regression line, given the dependent variables. For example, COGS may be estimated using the regression formula:

$$COGS_i = \alpha + \beta(Revenue_i) + \varepsilon_i \quad (2)$$

The conditional return distribution for COGS is a normal distribution with conditional mean,  $E(COGS_i | Revenue_i) = \alpha + \beta(Revenue_i)$ , and standard deviation,  $\sigma_\varepsilon$ .

There are several alternative methods to elicit subjective conditional probability distributions. Spetzler and Staël von Holstein (1975), Howard (1989), and Kemmerer et al. (2001) all describe methods to help elicit subjective conditional probability distributions. In our extended example below, we use both regression and subjective conditional probability distributions.

Another way to determine the numerical inputs for the network is through simulation. A spreadsheet simulation model can be used to determine the probabilities and state values for the firm-specific variables. Simulation software such as @Risk or Crystal Ball can be used in a spreadsheet model to generate a probability distribution for earnings per share. An analyst can use historical data on income statement items like revenue, gross margin,

and SG&A (selling, general, and administrative expenses) to estimate probability distributions for these variables.<sup>3</sup> One advantage of this method is that the analyst can incorporate assumptions for any items in the financial statements, and link the probability distribution to nodes in the Bayesian network.

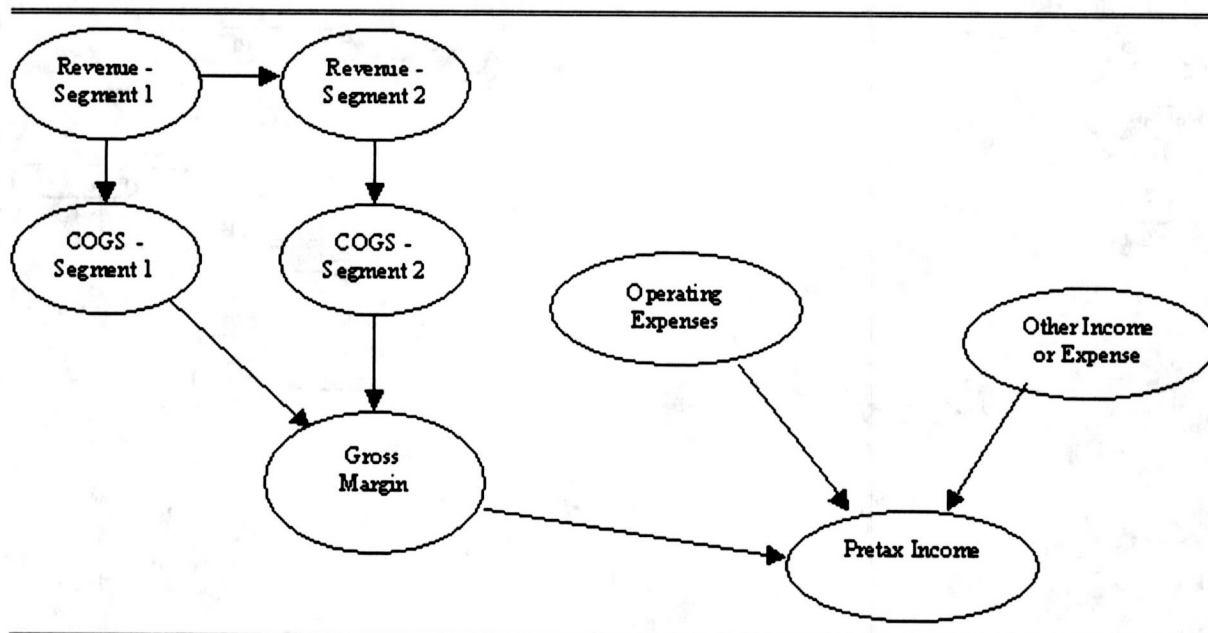
### C. Model Diagnostics

Since there are two parts to a Bayesian network, i.e., a graphical model and conditional probability tables for each node in the graph, both parts of the model need to be assessed for validity. In some instances, common statistical tools can aid in assessing model validity, but some unique aspects associated with Bayesian networks require other diagnostic tools.

A common statistical test is the test of independence. Arrows between nodes denote dependence between variables. The absence of an arrow between nodes denotes conditional independence. Testing for conditional independence involves testing for independence between two nodes for each state of the mediating variables.

For small models (or portions of a larger one), traditional tests of independence are feasible, but for large models such an approach is not tractable. Two general approaches are used as alternative tests of model validity. The first type of tests is non-parametric goodness of

<sup>3</sup>One can fit a distribution or use normal distributions for these variables with means and standard deviations calculated using historical numbers from financial statements.

**Exhibit 3. Alternative Model for Pretax Income**

fit tests (see Swartz, 1999, and Spiegelhalter, Best, Carlin, and Linde, 2002). The second type is simulation methods specific to Bayesian networks; Marshall and Spiegelhalter (2003) provide a review.

### III. An Example

Our Bayesian network example combines several macroeconomic and firm-specific variables that an analyst might use in a top-down analysis of selected stocks in the biomedical device industry. The output of the network is the portfolio value in thousands of dollars. This model represents only one way to incorporate macro and micro variables into security analysis. We discuss the choice of variables used and the calculations for different probability distributions.

Exhibit 4 shows the example Bayesian network for four firms in the biomedical device industry. The ultimate goal is to model the probability distribution for the portfolio value in the lower right-hand corner. We use the software, Netica, to create the graphical model and compute the model outputs.

The upper left-hand corner shows four macroeconomic variables that affect market return. We choose these variables based on work by Burmeister and Wall (1986), Burmeister, Wall and Hamilton (1986), and Berry, Burmeister, and McElroy (1988). These authors test a multifactor return model using a default risk premium (*DP*), a maturity premium (*MP*), unexpected changes in inflation (*INF*), and unexpected changes in productivity

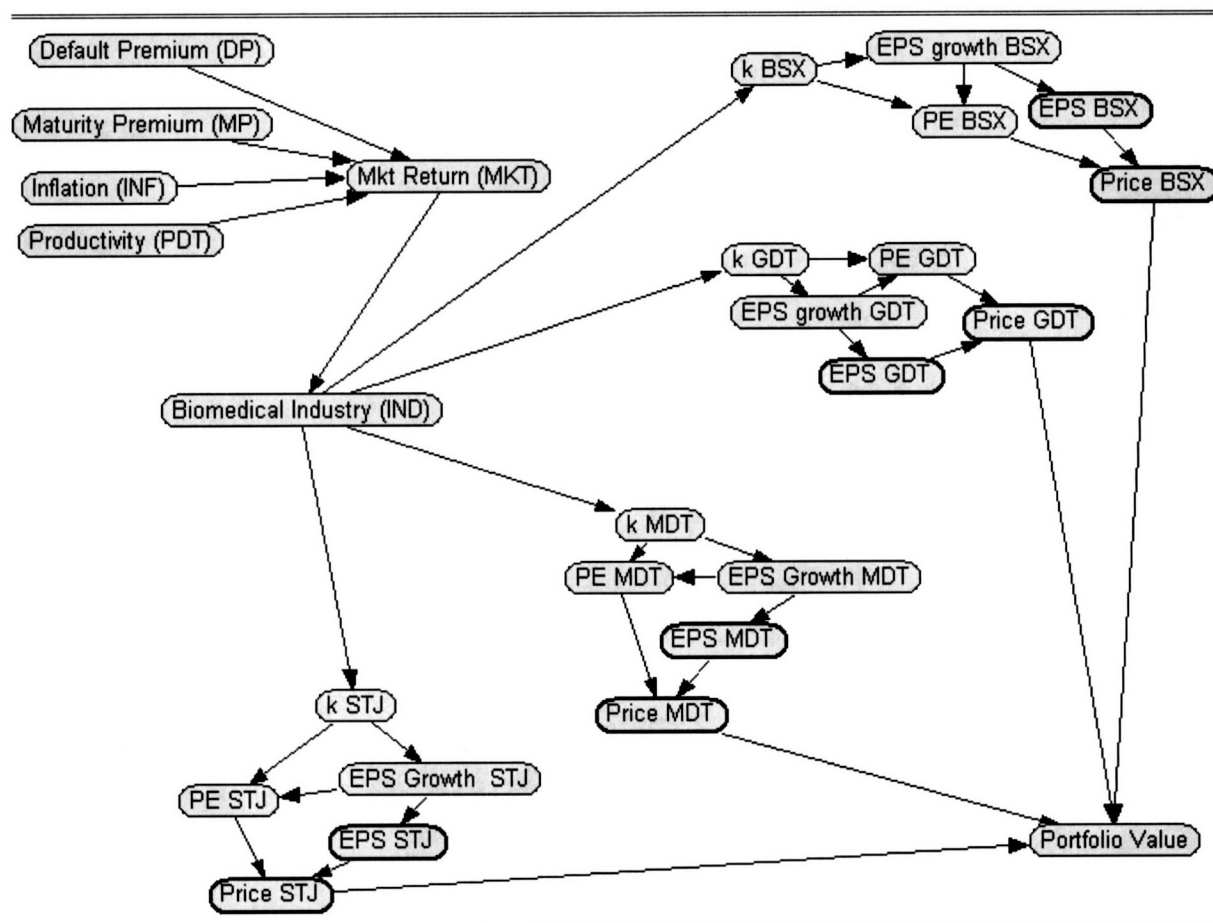
(*PDT*) to determine the expected market return (*MKT*). We condition the return of the biomedical industry (*IND*) on the market return. Construction of this Bayesian network assumes that the return on the biomedical industry is independent of macroeconomic factors, given the return on the market, consistent with the CAPM.

The first panel in Exhibit 5 describes the macroeconomic variables. This portion of the Bayesian network requires the estimation of five probability distributions. For each macroeconomic variable, we first categorize historical monthly observations over the period 1971 through 1999 into three states representing low, medium, and high values. We then use the bracket-median method of Clemen (1991) to fit a discrete probability distribution for the three states for each macroeconomic variable. We use the same technique to estimate the conditional probability distribution of market returns, given each state of the macroeconomic factors. This procedure can be regarded as a discretized version of a multifactor return model.

Note that the probability distributions in a Bayesian network do not have to be discrete; the analyst is free to make any assumption for the probability distributions in the model (continuous, fat tailed, truncated). Further detail on calculation of probability distributions is provided in the Appendix.

The four stocks in the biomedical device industry are: Guidant Corporation (*GDT*), St. John's Medical (*STJ*), Boston Scientific (*BSX*), and Medtronic, Inc. (*MDT*). For each stock, we model the required rate of return distribution, *k*, as a conditional probability distribution

## Exhibit 4. Example Biomedical Device Industry Bayesian Network



conditioned on the industry return distribution.<sup>4</sup> For this purpose, we use monthly historical observations on each stock as well as industry returns, and apply the same discretization technique to estimate a conditional probability distribution for each stock's return conditioned on the industry return.

The resulting Bayesian network assumes that the required return on a particular stock is independent of market returns, given the state of industry returns. This independence argument simplifies specification of the joint probability distribution of the variables in the model decomposed as:

$$P(k_{company}|IND) \times P(IND|MKT) \times P(MKT|DP,MP,INF,PDT) \times P(DP) \times P(MP) \times P(INF) \times P(PDT) \quad (3)$$

where  $P(k_{company}|IND)$  represents the conditional distribution of company returns conditioned on the state of

industry returns;  $P(IND|MKT)$  represents the conditional distribution of industry returns conditioned on market returns; and  $P(MKT|DP,MP,INF,PDT)$  represents the conditional distribution of market returns conditioned on the state of macroeconomic factors. The conditional independence assumption in the model asserts that  $P(k_{company}|IND) = P(k_{company}|IND,MKT)$ .

The probability distributions for macroeconomic variables,  $P(DP)$ ,  $P(MP)$ ,  $P(INF)$  and  $P(PDT)$ , are all marginal distributions. All probability distributions in the example are estimated using monthly historical observations.

Note that this probability model assumes that returns for each stock are conditionally dependent, given the state of the industry. In other words, given evidence on the state of the industry, the return distributions for the companies in the industry become dependent, indicating a certain degree of correlation between stocks within the industry. Company returns, however, are independent of all other economic and market variables, given the state of the industry.

Once again, the specific assumptions implicit in the network depend on how the modeler views the relations

<sup>4</sup>For example, the node labeled,  $k_{GDT}$ , is the required rate of return for Guidant (GDT).



## Exhibit 5. Description and Construction of Economic Firm Variables

| Variable                                           | Description                                                                                                            | Probability Distribution                                                     | Data Source                                                  |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Macroeconomic and Industry Variables</b>        |                                                                                                                        |                                                                              |                                                              |
| Default Risk Premium ( <i>DP</i> )                 | Long-term corporate bond return less long-term government bond return.                                                 | Marginal                                                                     | Ibbotson and Associates                                      |
| Maturity Premium ( <i>MP</i> )                     | Long-term corporate bond return less one-month treasury bill return                                                    | Marginal                                                                     | Ibbotson and Associates                                      |
| Unexpected Change in Inflation ( <i>INF</i> )      | Expected inflation at the beginning of the month less actual inflation at the end of the month                         | Marginal                                                                     | Ibbotson and Associates                                      |
| Unexpected Change in Productivity ( <i>PDT</i> )   | Expected growth in real final sales at the beginning of the month less actual real final sales at the end of the month | Marginal                                                                     | National Income Accounts                                     |
| Other Market Effects ( <i>MKT</i> )                | Expected market return using the 4 factors less actual market return                                                   | Conditional on variables { <i>DP</i> , <i>MP</i> , <i>INF</i> , <i>PDT</i> } | S&P 500 Index                                                |
| Industry effects ( <i>IND</i> )                    | Return on equally weighted biomedical supply index                                                                     | Conditional on variable { <i>MKT</i> }                                       | Constructed based on returns for six biomedical supply firms |
| <b>Firm-Specific Variables (for each security)</b> |                                                                                                                        |                                                                              |                                                              |
| <i>k</i> _Security                                 | Required return on security                                                                                            | Conditional on { <i>IND</i> }                                                | Historical security returns                                  |
| Growth_security                                    | Growth in earnings                                                                                                     | Conditional on { <i>k</i> }                                                  |                                                              |
| EPS_security                                       | Earnings per share                                                                                                     | Deterministic, $EPS_0(1 + \text{growth})$                                    |                                                              |
| PE_security                                        | Price-to-earnings ratio                                                                                                | Conditional on { <i>k</i> , growth}                                          |                                                              |
| Price_security                                     | Price                                                                                                                  | Deterministic, $PE \times EPS$                                               |                                                              |
| Portfolio Value                                    |                                                                                                                        | Deterministic, weighted average of security values                           |                                                              |

among the variables in the model. These conditional independence assumptions make it easier and more efficient to update all probabilities and portfolio value when evidence is added to the model.

The second panel of Exhibit 5 describes the firm-specific variables in the model. Further details on probability calculations are provided in the Appendix.

We next determine the P/E value for each company using a modification of the constant dividend growth rate model as  $PE\ ratio = \frac{1}{k - g}$ , where *k* is the required

rate of return, and *g* is earnings growth. We assume all earnings will be paid to shareholders in either the form of dividends or an equivalent capital appreciation. This is indicated in the network by two arcs entering into the node *PE\_company*. The two arcs indicate that the P/E ratio is based on earnings per share growth (*EPS\_growth\_company*) and the required rate of return (*k\_company*). For EPS growth we use a subjective probability distribution. We use a variety of analyst estimates for EPS growth rate, and construct a discrete probability distribution for this variable, given different analyst expectations.

The analyst is free to specify any subjective probability distribution for earnings growth; and here is where subjective judgments enter in the model. We then calculate the EPS value (*EPS\_company*) based on the most recent EPS value and the earnings growth rate. Finally, a

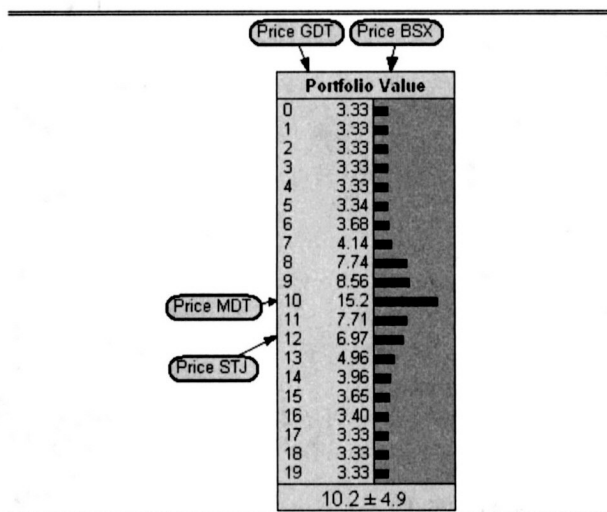
probability distribution is estimated for the stock price by simply extending two arcs into the node *Price\_company* where price is simply EPS times the P/E ratio. The last node in the model is *Portfolio Value* which is price times number of shares for each stock. For simplicity, we assume there are 100 shares of each stock, although calculation of the optimal number can easily be incorporated into the model. We use Netica to compute the probability distribution for portfolio value, given all the probability distributions and other numerical relations.

Exhibit 6 shows the extended view of the prior probability distribution of *Portfolio Value*. The numbers on the left show the state values. The first value, 0, represents the range of portfolio values from 0 to \$1,000, and the last value, 19, represents the range of portfolio values above \$19,000. The next column represents the probability of each state, given the price distributions of the four stocks and the number of shares owned. The expected value of the portfolio is \$10,200 with a standard deviation of \$4,900. The value-at-risk (VaR) at the 0.03 level is between 0 and \$1,000; at the 0.05 level it is between \$1,000 and \$2,000.

An important modeling decision is whether to use discrete probability models or a combination of both discrete and continuous probability models for the variables. Continuous distributions are normally used to estimate economic data and returns; their major drawback is that they complicate the process of adding new evidence.



### Exhibit 6. Prior Probability for Portfolio Value Nobe



Evidence is most naturally added in a discrete model, even when the probability distribution is continuous.

An example is the response to a question on expectations about inflation. An analyst is most likely to give a point estimate or a general estimate such as "higher than normal." We chose to model the economic variables as discrete variables for two reasons. First, inferences based on the graphical structure of the model are relatively robust to the form of the inputs (Henrion et al., 1994). Second, and most important, we want to model the effect of new evidence on stock prices or portfolio value, so we decided to make the model more amenable to the addition of evidence.

## IV. Adding Evidence

An important feature of the Bayesian network framework is that it allows the decision maker to add new qualitative or quantitative information as evidence. Our model lets the user add evidence in several ways. First, the decision maker can use expert judgment on any variables to update the probabilities assigned to certain states of those variables. This is the traditional form of adding evidence to a Bayesian network. Other ways include updates related to events that affect particular companies or industries and updates to reflect real structural changes in the values of the variables.

*Evidence* can be defined as any information that changes the probability distribution in the network. An analyst who is bullish in expectations about the market overall, for example, can respecify the probabilities assigned to specified states for the node representing market return (*MKT*). The model will then recalculate all related conditional probabilities that go into final portfolio value.

Another example would be a news report related to a company or industry. On July 10, 2001, Guidant failed to win Food and Drug Administration backing for its pacemaker-like heart failure device. Given this information, we assume that earnings for the company would be negatively affected. Therefore, we add evidence to the Bayesian network that earnings growth for GDT would be in the low state.

Exhibit 7 shows the GDT and portfolio probability distributions before adding evidence, and Exhibit 8 shows the same view with the addition of the new evidence.

Exhibit 7 indicates a prior expected Guidant stock price of \$19.40. Incorporating the new evidence into the model produces a posterior expected stock price of \$15.90, which corresponds to approximately 18% price decline. The actual market reaction to this announcement was a 13.75% decline from the close on July 9, 2001, to the close on July 10. With the announcement made at 12:37 p.m. on July 10 (*Bloomberg News*), the total decline through July 11 was 23%.

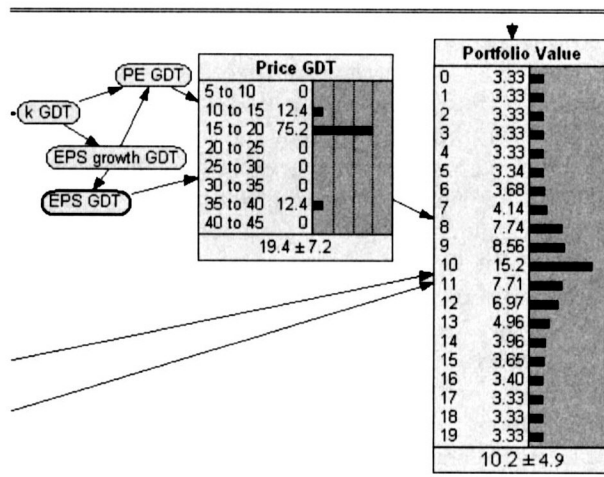
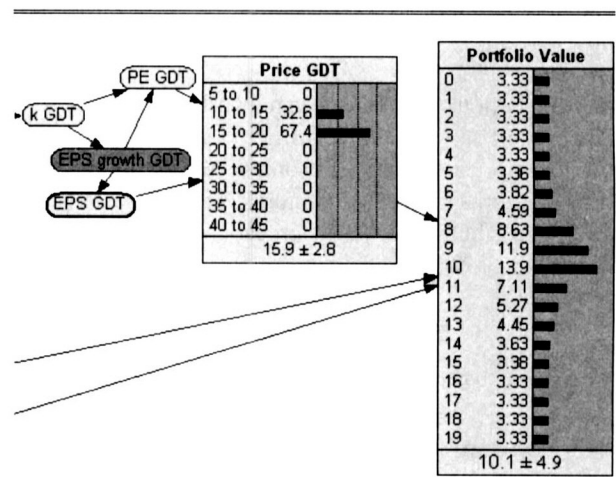
Evidence can also be added to update the actual values of states of variables. In our example, we rely on historical data to identify the low, medium, and high ranges of chance variables. In the case of macroeconomic factors, we base the range of values on historical data for inflation, the maturity premium, and so on. When an analyst believes a structural change has occurred in the probability distributions for these variables, different value ranges can be specified.<sup>5</sup>

In Exhibit 8, the VaR for the portfolio has not changed, even though the probability of some lower states has increased and the probability of some higher states has fallen. For example, the probability of the portfolio value being between \$10,000 and \$11,000 has declined from 0.152 to 0.139. The very lowest states between 0 to \$4,000 have not changed, but the \$5,000 state has become slightly more likely.

## V. Limitations, Extensions, and Conclusion

A major challenge in security analysis is incorporating qualitative information in an analysis in a systematic way. Analysts often combine historical data with qualitative, imprecise evidence. An analyst may consider, for example, a firm's management or the reliability of the financial statements. Although traditional pricing models in the finance literature deal with the 'average' firm, in practice, security analysts must assess a large amount of sometimes conflicting data to make a decision based on uncertain information about special situations and individual firms.

<sup>5</sup>Structural changes may be due to the expectation of a crash, a major default, or any other event that potentially leads to a shift in the probability distributions for the variables in the model.

**Exhibit 7. GDT and Portfolio Distribution Before Adding Evidence****Exhibit 8. GDT and Portfolio Distribution after Adding Evidence**

The uncertain, ambiguous, or sometimes, incomplete nature of the data faced by security analysts requires an approach that will combine the quantitative information with qualitative information in a systematic way.

We have proposed a decision support system for equity analysts that use a Bayesian network. Bayesian networks have been applied in a wide variety of decision-making uses such as venture capital financing, auditing, medical diagnosis, and software design. The model we propose provides a systematic approach for a decision maker to represent uncertain, ambiguous, incomplete, and conflicting information that security analysts and portfolio managers routinely analyze. It allows for incorporation of both quantitative data and qualitative data in the decision-making process and uses macroeconomic and firm-specific data to develop industry returns, equity valuations, and portfolio value.

The Bayesian network lets an analyst see the effect of new evidence on various outcomes, including portfolio value and risk. A Bayesian network can also provide feedback to an analyst to help improve learning and reduce bias in forecasts.

Our example demonstrates how to update the analysis as new information becomes available. The model is adaptable to various financial decision making contexts and different types of evidence. It can be extended for use in other industries and for diversified portfolios. An optimization module can be incorporated in an asset

allocation model to calculate optimal weights and revise them as evidence is added.

The Bayesian framework can be a risk management tool because the output is a probability distribution for the value of the portfolio. A logical extension is to incorporate value-at-risk as an additional tool for portfolio managers. Posterior expected return and standard deviation estimates can be easily estimated and used to revise VaR estimates after new evidence is added. The model can also be used in a simulation model to determine the VaR under various scenarios, such as extreme market conditions representing low-probability scenarios.

Bayesian networks should help analysts make unbiased decisions that are not based on emotional or behavioral aspects of investing but rather on conditional probabilities that incorporate quantitative and qualitative data in an explicit manner. We thus can hope analysis may overcome the vagaries of the market. As Graham (1973, p.107) observes:

*The investor who permits himself to be stampeded or unduly worried by unjustified market declines in his holdings is perversely transforming his basic advantage into a basic disadvantage. That man would be better off if his stocks had no market quotation at all, for he would then be spared the mental anguish caused him by other persons' mistakes of judgment. ■*

## Appendix: Calculation Details for the Bayesian Network

Inputs into the Bayesian network are either macroeconomic or firm-specific variables.

### A. Macroeconomic Inputs

Following Berry, Burmeister, and McElroy (1988), we assume that the probability distribution for market

return (*MKT*) is a function of macroeconomic factors including the default risk premium (*DP*), the maturity premium (*MP*), unexpected changes in inflation (*INF*), and unexpected changes in productivity (*PDT*). All variables are measured as monthly percentages from 1971 through 1999.

This portion of the Bayesian network requires the construction of six probability distributions, a marginal distribution for each factor and conditional distributions for market returns and industry returns. Thus, the joint probability distribution for the macroeconomic portion of the network is decomposed as

$$P(DP) \times P(MP) \times P(INF) \times P(PDT) \\ \times P(MKT | DP, MP, INF, PDT) \quad (4)$$

The joint probability distribution for market and industry returns is decomposed as

$$P(MKT) \times P(IND | MKT) \quad (5)$$

Several descriptive statistics of the macroeconomic factors from 1971 through 1999 are provided in Figure 1.

The historical monthly observations on each of these factors are discretized using the bracket-median method as low, medium, or high. Following McNamee and Celona (1987), we assign interval probabilities 0.25, 0.50, and 0.25 to the states rather than the equally likely intervals usually used with the bracket-median method.<sup>6</sup>

Figure 2 provides the fitted marginal probability distributions and corresponding ranges of factor values for each state (low, medium, high) obtained from monthly historical observations.

The low state for the variable default premium (*DP*) corresponds to a range of values from -3.96% through -1.22% at an estimated probability of 0.25. The analyst assigns subjective probabilities to each factor range depending on expectations about these factors.

We then use monthly returns for the market (the S&P 500 index) and the industry to estimate conditional probability distributions for market returns given economic factors,  $P(MKT|DP, MP, INF, PDT)$ , and industry returns, given market returns,  $P(IND|MKT)$ . For example the conditional probability of *MKT*, given variables  $\{DP, MP, INF, PDT\}$  has 81 ( $3^4$ ) possible states as each of the conditioning variables has three possible states.

Figure 3 shows a sample portion of the conditional probabilities for the variable *MKT*.<sup>7</sup> The numbers represent the frequency of market returns, given each combination of states of macroeconomic factors. For example, the probability  $P(MKT = \text{low} | DP = \text{low}, MP = \text{low}, INF = \text{low}, PDT = \text{low}) = 0.4$  indicates that 40% of monthly observations of market returns during 1971-1999 lie in the low range for market returns, given that the macroeconomic factors are also in the low range. So, the conditional probabilities are simply observed frequencies of historical returns conditioned on ranges of values for macro factors.

Note that the estimated conditional probability for high market returns, given that macro factors are all in the low state, is zero, indicating that such a combination is not observed in the sample period. The ranges of market returns are [-3.92; 0.96], [0.96; 4.29], and [4.29; 16.3] (all in percentages) for low, medium, and high states by the bracket-median method.

Figure 4 provides the conditional probability distribution of biomedical industry returns (*IND*), given the return on the market (*MKT*). The ranges are estimated as [-21.76; 1.40], [1.40; 19.89], and [19.89; 35.20] (all in percentages) for low, medium and high states. There are a wide range of returns for each state because of high variability of returns in the highly risky biomedical industry. Here again, analysts can rely on subjective judgment for the specification of each state; our estimations are based on historical observations.

Conditional probabilities are then estimated for each state of industry returns, given market returns. The probability  $P(IND = \text{Low} | MKT = \text{Low}) = 0.46$  in Exhibit A-4 indicates that 46% of the observations on industry returns fall into the low return category when market return is also in the low category.

## B. Firm-Specific Inputs

The probabilistic model relating the macroeconomic factors to the required rate of return for the company can be regarded as the discretized version of the multi-factor return models. We construct the firm-specific portion of the network as another conditional probability model. For each firm, four variables are specified representing EPS growth, price-earnings ratio (PE), expected EPS, and expected price. EPS growth is estimated using a subjective probability distribution.

Typically, higher growth companies have higher required rates of return. Given the expected return for

<sup>6</sup> In the bracket-median method, one divides the continuous cumulative probability distribution into  $n$  equally probable intervals. In the discrete distribution, each interval is represented by the point corresponding to the median of that interval, and each point is assigned probability  $1/n$ .

<sup>7</sup> Our method is a discretization of the method used in the Berry et al. (1988) multifactor model. In the continuous case, regression provides estimates of the conditional mean and standard deviation, assuming that a variable is normally distributed with the estimated conditional parameters.

**Figure 1. Descriptive Statistics for Macroeconomic Factors**

|                | DP    | MP    | INF   | PDT    |
|----------------|-------|-------|-------|--------|
| Average (%)    | 0.01  | 0.21  | -0.01 | 0.37   |
| Std. Deviation | 1.11  | 3.03  | 0.17  | 3.55   |
| Median         | -0.06 | 0.11  | -0.01 | 0.29   |
| Minimum        | -3.96 | -9.28 | -0.56 | -20.10 |
| Maximum        | 5.14  | 13.97 | 0.78  | 22.92  |

**Figure 2. Marginal Distributions for Macroeconomic Factors**

| State  | Probability | DP (%)         | MP (%)         | INF (%)        | PDT (%)         |
|--------|-------------|----------------|----------------|----------------|-----------------|
| Low    | 0.25        | [-3.96; -1.22] | [-9.28; -3.44] | [-0.55; -0.21] | [-20.09; -3.53] |
| Medium | 0.50        | [-1.22; 1.36]  | [-3.44; 3.93]  | [-0.21; 0.21]  | [-3.53; 4.37]   |
| High   | 0.25        | [1.36; 5.14]   | [3.93; 13.97]  | [0.21; 0.77]   | [4.37; 22.91]   |

**Figure 3. Conditional Probabilities for Market Return, Given Macroeconomic Factors**

| States for {DP, MP, INF, PDT} |     |     |        | P(MKT DP, MP, INF, PDT) |        |      |
|-------------------------------|-----|-----|--------|-------------------------|--------|------|
| DP                            | MP  | INF | PDT    | Low                     | Medium | High |
| Low                           | Low | Low | Low    | 0.40                    | 0.60   | 0.00 |
| Low                           | Low | Low | Medium | 0.50                    | 0.20   | 0.30 |
| Low                           | Low | Low | High   | 0.60                    | 0.20   | 0.20 |

each stock, we calculate the expected growth rates as  $P(\text{Growth}) = P(k) \times P(\text{Growth}|k)$ . Figure 5 presents the conditional probability table we use for all stocks.

We then formulate the P/E ratio for each company using a modification of the constant dividend growth rate model,  $PE \text{ ratio} = \frac{1}{k - g}$ , where  $k$  is the required rate of return, and  $g$  is earnings growth.

We assume all earnings will be returned to shareholders either in the form of dividends or capital appreciation. Thus, the price earnings ratio in the network has a conditional probability distribution based on earnings growth and the required rate of return on the stock, indicated by two arcs entering into the node  $PE\_company$ . We then formulate the expected EPS value ( $EPS\_company$ ) based on the most recent EPS value ( $EPS_0$ ) and the earnings growth rate ( $g$ ) as:

$$EPS\_company = EPS_0 \times (1 + g).$$

We can use BSX to illustrate the relations between expected growth and the P/E ratio. The previous period earnings per share for BSX were 0.905. Analysts' esti-

mates for the next year were obtained from First Call estimates. We then classify these estimates into low, medium, or high states as shown in Figure 6.

The implied earnings growth rates using the EPS estimates in Figure 6 are -1% for the low state, 4% for the medium state, and 7% for the high state. We specify the values of BSX expected returns for each state as 8% in the low state, 10% in the medium state and 12% in the high state. This produces the P/E ratios Figure 7 for each state. For example, when  $k$  is 12% and  $g$  is -1%, the P/E ratio is 8. Since that is the lowest P/E ratio obtained when  $k$  is in the high state and  $g$  is in the low state, we assign a probability of 1 to low P/E.

Since the relation between expected EPS and prior period EPS is deterministic given the growth rate, we can fully determine EPS, given  $g$ , using  $EPS\_company = EPS_0 \times (1 + g)$ .

Finally, a probability distribution is estimated for the stock price by extending two arcs into the node representing a deterministic function as

$$Price\_company = EPS\_company \times PE\_company \quad (6)$$

The last step is to use the price estimates for each individual firm to calculate a probability distribution on the

**Figure 4. Conditional Probabilities for Biomedical Industry Returns**

| State of <i>MKT</i> | Biomedical Industry Returns, $P(IND MKT)$ |        |      |
|---------------------|-------------------------------------------|--------|------|
|                     | Low                                       | Medium | High |
| Low                 | 0.46                                      | 0.50   | 0.04 |
| Medium              | 0.16                                      | 0.57   | 0.27 |
| High                | 0.21                                      | 0.36   | 0.43 |

**Figure 5. Conditional Probabilities for Earnings Growth Rates**

| State of <i>k</i> | Growth, $P(Growth k)$ |        |      |
|-------------------|-----------------------|--------|------|
|                   | Low                   | Medium | High |
| Low               | 0.75                  | 0.25   | 0.00 |
| Medium            | 0.10                  | 0.80   | 0.10 |
| High              | 0.00                  | 0.25   | 0.75 |

**Figure 6. First Call Estimates for BSX Earnings per Share**

| EPS  | Number of Analysts | State  | EPS  | Number of Analysts | State |
|------|--------------------|--------|------|--------------------|-------|
| 0.84 | 1                  | Low    | 0.98 | 3                  | High  |
| 0.89 | 2                  | Low    | 1.00 | 3                  | High  |
| 0.90 | 1                  | Low    | 1.02 | 1                  | High  |
| 0.92 | 1                  | Medium | 1.03 | 1                  | High  |
| 0.93 | 2                  | Medium | 1.05 | 1                  | High  |
| 0.94 | 1                  | Medium | 1.06 | 2                  | High  |
| 0.95 | 2                  | Medium | 1.07 | 1                  | High  |

**Figure 7. Conditional Probabilities for BSX P/E Growth and Expected Return**

| Values   |          | States   |          | Value | State    | $P(P/E\_BSX k\_BSX,g\_BSX)$ |        |      |
|----------|----------|----------|----------|-------|----------|-----------------------------|--------|------|
| <i>k</i> | <i>g</i> | <i>k</i> | <i>g</i> | P/E   | P/E      | Low                         | Medium | High |
| 0.12     | -0.01    | High     | Low      | 8     | Low      | 1.00                        | 0.00   | 0.00 |
| 0.10     | -0.01    | Medium   | Low      | 9     | Low      | 0.75                        | 0.25   | 0.00 |
| 0.08     | -0.01    | Low      | Low      | 11    | Low/Med  | 0.25                        | 0.75   | 0.00 |
| 0.12     | 0.04     | High     | Medium   | 13    | Low/Med  | 0.50                        | 0.50   | 0.00 |
| 0.10     | 0.04     | Medium   | Medium   | 17    | Med/High | 0.00                        | 0.25   | 0.75 |
| 0.12     | 0.07     | High     | High     | 20    | Med/High | 0.00                        | 0.75   | 0.25 |
| 0.08     | 0.04     | Low      | Medium   | 25    | High     | 0.00                        | 0.50   | 0.50 |
| 0.10     | 0.07     | Medium   | High     | 33    | High     | 0.00                        | 0.25   | 0.75 |
| 0.08     | 0.07     | Low      | High     | 100   | High     | 0.00                        | 0.00   | 1.00 |

portfolio value. Four arcs entering the *Portfolio Value* node specify another deterministic function as

$$\text{Portfolio Value} = \sum_{\text{company}=1}^4 \text{Price\_company} \times n_{\text{company}} \quad (7)$$

For simplicity, we assume that the portfolio holds 100 shares of each stock and define value ranges in thousands of dollars for each possible state of portfolio value. This fully specifies the probability distribution for portfolio value.

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